BRNO UNIVERSITY OF TECHNOLOGY FACULTY OF ELECTRICAL ENGINEERING AND COMMUNICATION

Robotics and AI

MPC-MAP Assignment No. 3 - Report

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Task 1

The map syntax uses coordinates line by line: goal, map boundaries, walls, and GNSS denied zone after the inf keyword. I wanted my custom map to be big and simple for easier testing in subsequent tasks. The map is defined in *maps/custom_1.txt*.

Task 2

I created three custom paths in the function *algorithms/path_planning/custom_path.m*. The paths are as recommended: line, sine, arc. The line will be good for PID tuning and sine for testing sharp turns.

Task 3

First, I wanted to implement an algorithm based on XTE error tracking. But the PID tuning was a little complex. I tried to set the gains by the Ziegler–Nichols method, but the turns in the sine path were really imprecise, the example run with Ziegler–Nichols is in the Appendix in Figure 2. So, I edited the parameters based on testing. Finally, I also had to adjust the forward velocity, so the robot runs at maximum speed only when the XTE is small.
Second, I tried to implement a pure pursuit algorithm. There was no need for PID, so the implementation was much simpler. Also, I tried different values of the look-ahead distance L, and in this application the optimal value is somewhere between 0,2 m and 0,5 m. The exact value depends on the path shape and map. Smaller value means more oscillation and a bigger value means a higher cross-track error.

In the comparison of these two methods, the robot reached the goal in a very similar number of iterations. Then I compared the cross-track error between the two methods, the results are in Figure 1.

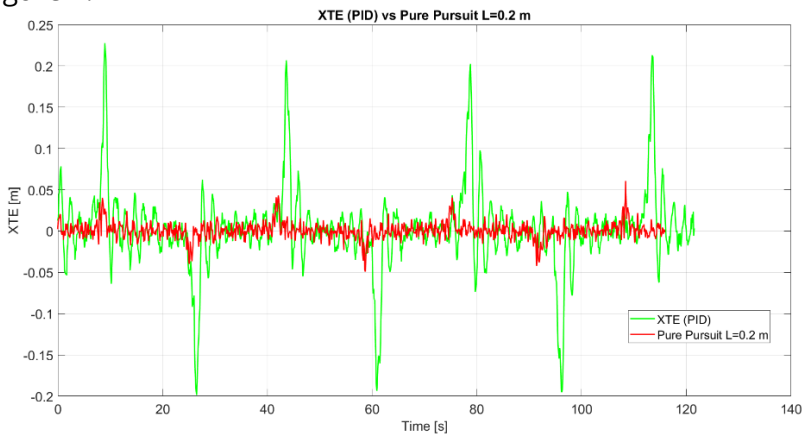

Fig. 1 – Cross track error between methods (sine path)

As shown in Figure 1, I could not reach better accuracy with the XTE method, and I reached better results with pure pursuit. This could be caused by the imprecisely tuned PID controller.

Appendix

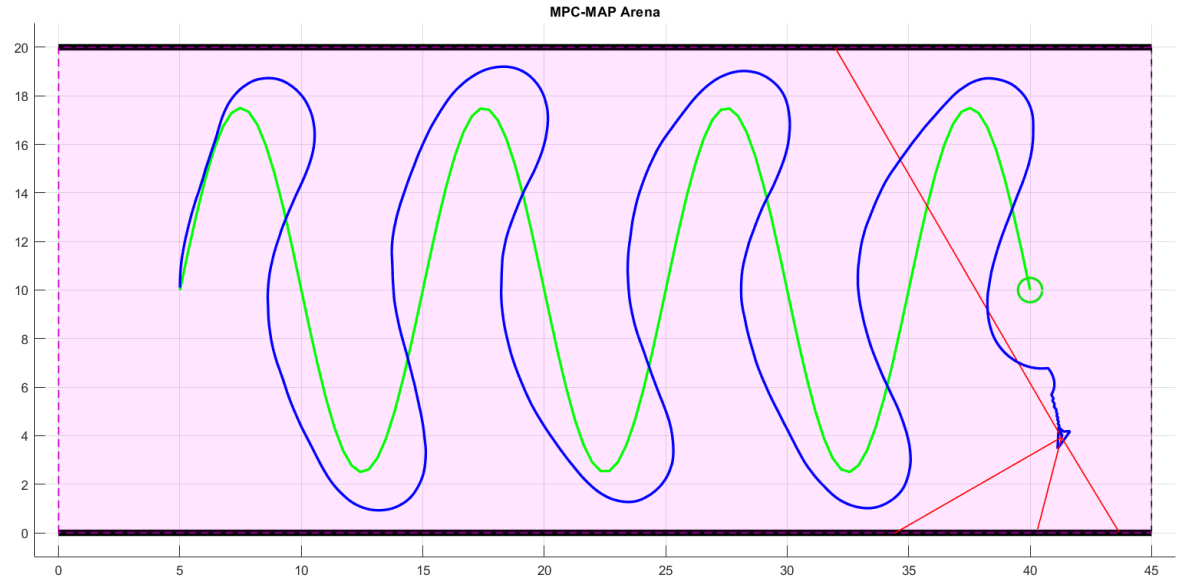

Fig. 2 – XTE method after Ziegler-Nichols PID tunning

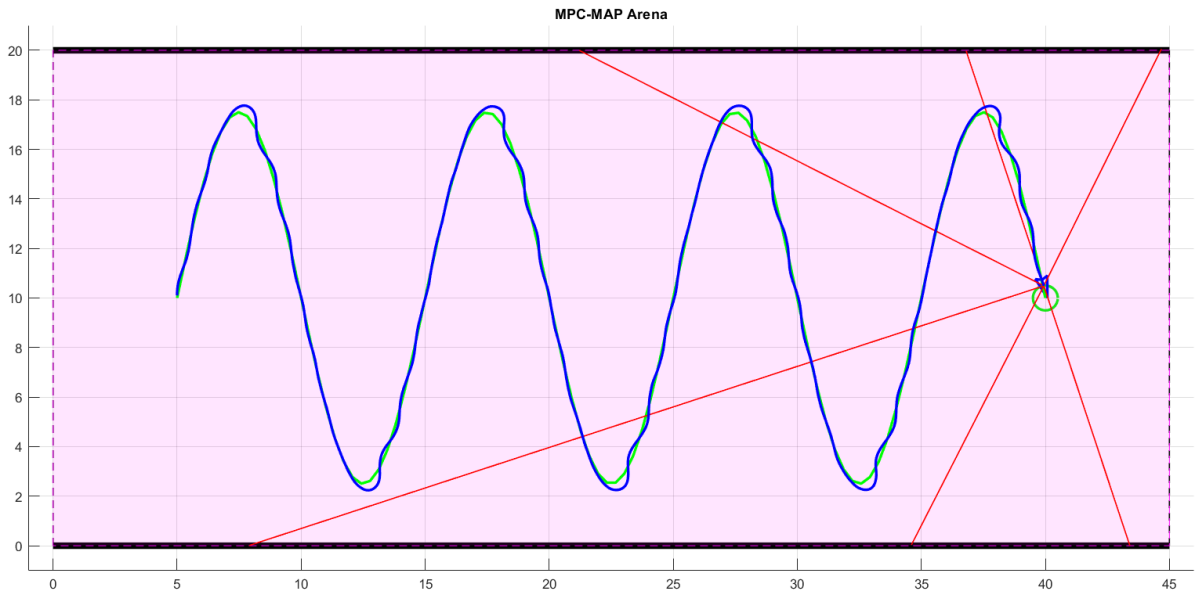

Fig. 3 – XTE method after edit Ziegler-Nichols gains

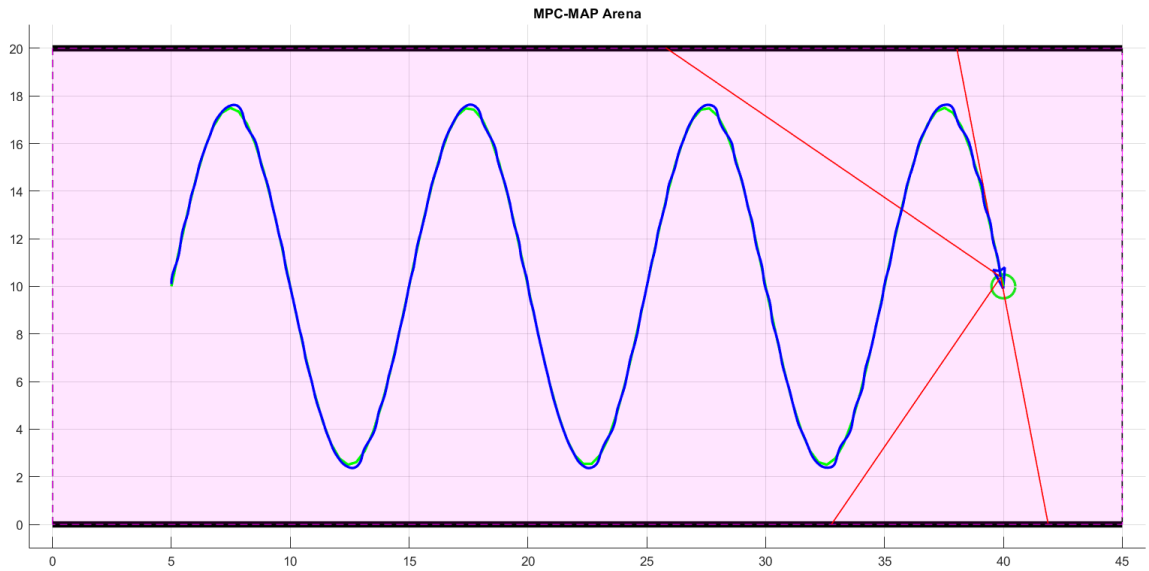

Fig. 4 – XTE method after adjusting forward velocity based on XTE

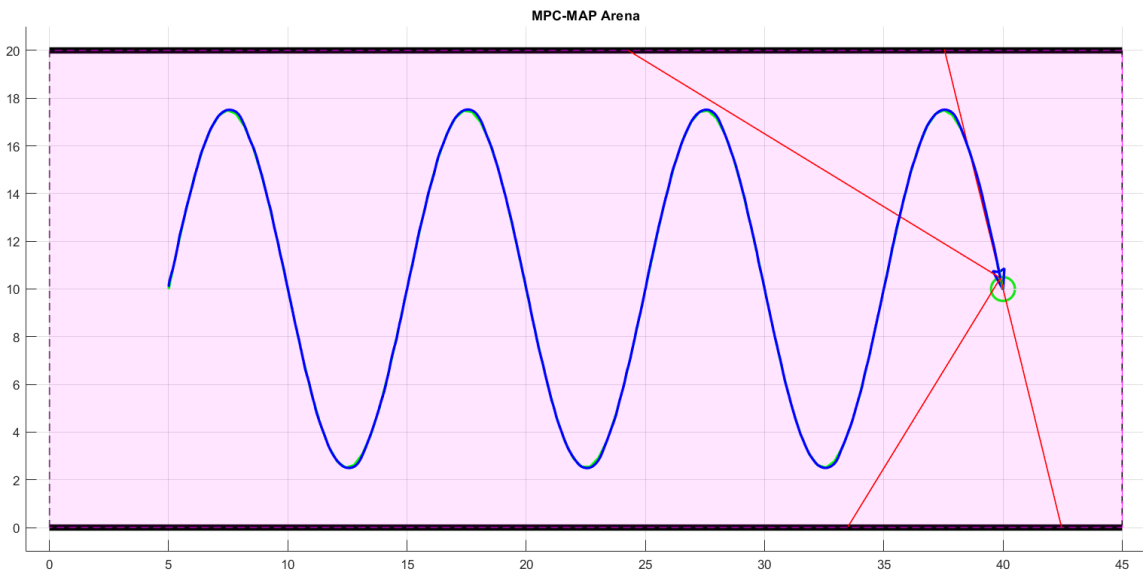

Fig. 5 – Pure pursuit method L=0,3 m